

Courses of Algebra 1

Chapter 1

Logic and methods of proof

1.1 Propositions

Definition 1.1 *A proposition is a statement that is either true (we note 1 or T) or false (we note 0 or F), but not both. Propositions generally noted by the letter $P; Q; R; S; \dots$*

Examples

1. P : " $9 > 7$ ". This statement is true, therefore it is a proposition.
2. Q : " 4 is a prime number". This statement is false, so it is a proposition.
3. R : " $x + 1 > 0$ ". This statement is neither true nor false, so it is not a proposition.

Definition 1.2 *Any proposition proven true is called a Theorem (for example the Theorem of PYTHAGORES, Thales...).*

Definition 1.3 (*Truth table*) *We summarize all the possibilities of the truth values of propositions in a table called a truth table.*

- i. For a proposition there are two possibilities

P
1
0

Truth table of proposition P

ii. For two propositions there are four possibilities

P	Q
1	1
1	0
0	1
0	0

Truth table of two propositions.

iii. For n propositions there are 2^n possibilities.

1.2 Logical connectors

Logical connectors are operations to form a new proposition.

I. Negation (No P , $\bar{P}$)

Definition 1.4 *The negation of a proposition P is the proposition denoted $\bar{P}$ is true if P is false and is false if P is true.*

$\bar{P}$ reads: No P .

P	$\bar{P}$
1	0
0	1

Truth table for the connector "No P ".

Examples

1. P : " $2 > 1$ ", $\bar{P}$: " $2 \leq 1$ ".
2. Q : "the function f is increasing", $\bar{Q}$: "the function f is not increasing".
3. R : "2 is a positive number", $\bar{R}$: "2 is a negative number".

II. The conjunction ($P \wedge Q$)

Definition 1.5 *The conjunction of the two propositions P and Q is the proposition denoted $P \wedge Q$. The conjunction $P \wedge Q$ is true when both P and Q are true, and is false otherwise.*

$P \wedge Q$ reads: P and Q .

P	Q	$P \wedge Q$
1	1	1
1	0	0
0	1	0
0	0	0

Truth table for the connector " $\wedge$ ".

Examples

1. $\underbrace{(2 + 1 = 3)}_P \wedge \underbrace{(1 = 1)}_Q$ this proposition is true because P is true and Q is true. (1st row of the table).
3. $\underbrace{(1 < 0)}_P \wedge \underbrace{(3 > 0)}_Q$ this proposition is false because P is false and Q is true (3th row of the table)
4. $\underbrace{(1 < 0)}_P \wedge \underbrace{(-1 > 0)}_Q$ this proposition is false because P is false and Q is false (4th row of the table).

III. The disjunction ($P \vee Q$)

Definition 1.6 The disjunction of P on Q is the proposition denoted by $P \vee Q$. The disjunction $P \vee Q$ is false when both P and Q are false, and is true otherwise

$P \vee Q$ reads: P or Q .

P	Q	$P \vee Q$
1	1	1
1	0	1
0	1	1
0	0	0

Truth table for the connector " $\vee$ ".

Example

2. $\underbrace{(2 > 1)}_P \vee \underbrace{(2 < 0)}_Q$ this proposition is true because P is true and Q is false .

IV. The implication ($P \implies Q$)

Definition 1.7 The implication of the two propositions P then Q is the proposition denoted $P \implies Q$. The implication $P \implies Q$ is false when P is true and Q is false, and is true in the other cases.

$P \implies Q$ reads : P implied Q , or : If P then Q .

P	Q	$P \implies Q$
1	1	1
1	0	0
0	1	1
0	0	1

Truth table for the connector " $\implies$ ".

Examples

- $\underbrace{(1 < 2)}_P \implies \underbrace{(3 \leq 2)}_Q$, this proposition is false because P is true and Q is false.
- $\underbrace{(1 = 2 \wedge 3 = 2)}_P \implies \underbrace{4 = 4}_Q$, this proposition is true because P is false and Q is true.

Definition 1.8 The converse of the implication $P \implies Q$ is

$$Q \implies P.$$

The contrapositive of implication $P \implies Q$ is

$$\bar{Q} \implies \bar{P}.$$

Examples

- The converse of the implication $\underbrace{x \leq 4}_P \implies \underbrace{\sqrt{x} \leq 2}_Q$ is $\underbrace{\sqrt{x} \leq 2}_Q \implies \underbrace{x \leq 4}_P$.
- The contrapositive of implication $\underbrace{x \leq 4}_P \implies \underbrace{\sqrt{x} \leq 2}_Q$ is $\underbrace{\sqrt{x} > 2}_{\bar{Q}} \implies \underbrace{x > 4}_{\bar{P}}$.

V. Equivalence ($\iff$) (if and only if)

Definition 1.9 The equivalence of the two propositions P and Q is the proposition denoted by $P \iff Q$. The equivalence $P \iff Q$ is true when P and Q have the same truth value, and is false otherwise.

$P \iff Q$ reads : P equivalent to Q , or : P if and only if Q .

P	Q	$P \Leftrightarrow Q$
1	1	1
1	0	0
0	1	0
0	0	1

Truth table for the connector " $\Leftrightarrow$ ".

Example

1. $\underbrace{2 = 5 - 3}_P \Leftrightarrow \underbrace{5 = 2 - 3}_Q$, (this proposition is false).

Theorem 1.1 Let P, Q be two propositions, we have

$$\overline{(P \Rightarrow Q)} \Leftrightarrow (P \wedge \bar{Q}).$$

Examples

1. $\underbrace{(x \in [0, 1])}_P \Rightarrow \underbrace{(x \geq 0)}_Q$ its negation : $\underbrace{(x \in [0, 1])}_P \wedge \underbrace{(x < 0)}_{\bar{Q}}$.
2. $\underbrace{(n \text{ is odd})}_P \Rightarrow \underbrace{(n^2 \text{ is odd})}_Q$ its negation : $\underbrace{(n \text{ is odd})}_P \wedge \underbrace{(n^2 \text{ is even})}_{\bar{Q}}$.

Properties of logical connectors

Let P, Q and R be three propositions. Then the following propositions are true.

1. $\bar{\bar{P}} \Leftrightarrow P$
2. $\overline{(P \wedge Q)} \Leftrightarrow (\bar{P} \vee \bar{Q}),$
3. $\overline{(P \vee Q)} \Leftrightarrow (\bar{P} \wedge \bar{Q}),$
4. $(P \Rightarrow Q) \Leftrightarrow (\bar{P} \vee Q),$
5. $(P \Leftrightarrow Q) \Leftrightarrow (\bar{P} \Leftrightarrow \bar{Q}),$
6. $(P \Leftrightarrow Q) \Leftrightarrow [(P \Rightarrow Q) \wedge (Q \Rightarrow P)],$
7. $[P \Rightarrow Q \wedge Q \Rightarrow R] \Rightarrow (P \Rightarrow R),$
8. $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$ (the distribution of $\wedge$ on $\vee$),
9. $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$ (the distribution of $\vee$ on $\wedge$),
10. $(P \wedge Q) \Leftrightarrow (Q \wedge P),$
11. $(P \vee Q) \Leftrightarrow (Q \vee P),$

1.3 Logical quantifiers " $\forall$ ", " $\exists$ ", " $\exists!$ "

1- The universal quantifier " $\forall$ ".

$\forall x$: means that *for all* x or *for every* x .

Example

1. $\forall x \in \mathbb{R} : x^2 \geq 0$.

2- The existential quantifier " $\exists$ ".

$\exists x$: means *there exists* x .

Example

1. $\exists x \in \mathbb{R} : x^2 - 1 \geq 0$.

3- The unique existential quantifier " $\exists!$ "

$\exists!x$: means *there is a unique* x .

Example

$\exists!x \in \mathbb{R}_+ : x^2 - 1 = 0$.

The negation of quantifiers

Let E be a set and $P(x)$ a proposition, we have

1. The negation of $\forall x \in E : P(x)$ is $\exists x \in E : \overline{P(x)}$ i.e.

$$\overline{\forall x \in E : P(x)} \iff \exists x \in E : \overline{P(x)}.$$

2. The negation of $\exists x \in E : P(x)$ is $\forall x \in E : \overline{P(x)}$ i.e.

$$\overline{\exists x \in E : P(x)} \iff \forall x \in E : \overline{P(x)}.$$

Examples

1. $\overline{\forall x \in \mathbb{R} : x^2 \geq 0} \iff \exists x \in \mathbb{R} : x^2 < 0$.

2. $\overline{\exists x \in \mathbb{R} : f(x) = g(x)} \iff \forall x \in \mathbb{R} : f(x) \neq g(x)$.

Remark 1.1 *The negation of $\exists!x \in E : P(x)$ is*

$$\overline{\exists!x \in E : P(x)} \iff \left[\left(\forall x \in E : \overline{P(x)} \right) \vee \left(\exists x, y \in E : P(x) \wedge P(y) \right) \right].$$

Example

$$\overline{\exists!x \in \mathbb{N} : x^2 - 1 = 0} \iff (\forall x \in \mathbb{N} : x^2 - 1 \neq 0) \vee (\exists x, y \in \mathbb{N} : x^2 - 1 = 0 \wedge y^2 - 1 = 0).$$

The order of quantifiers

Let E be a set and $P(x, y)$ a proposition, we have

$$1) \forall x \in E, \forall y \in E : P(x, y) \iff \forall y \in E, \forall x \in E : P(x, y).$$

$$2) \exists x \in E, \exists y \in E : P(x, y) \iff \exists y \in E, \exists x \in E : P(x, y).$$

Remark 1.2 The propositions $\forall x \in E, \exists y \in E : P(x, y)$ and $\exists y \in E, \forall x \in E : P(x, y)$ are different, in the first y depends on x while in the second y does not depend on x .

$$\forall x \in E, \exists y \in E : P(x, y) \not\iff \exists y \in E, \forall x \in E : P(x, y).$$

Example $\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : x + y \geq 0$ this proposition is true. But the proposition $\exists x \in \mathbb{R}, \forall y \in \mathbb{R} : x + y \geq 0$ is false. So

$$\forall x \in \mathbb{R}, \exists y \in \mathbb{R} : x + y \geq 0 \text{ not equivalent to } \exists x \in \mathbb{R}, \forall y \in \mathbb{R} : x + y \geq 0.$$

1.4 Methods of proof

1.4.1 Proof by truth table

To demonstrate that a proposition is valid, we check that its truth table contains only “1”.

To demonstrate that two propositions are equivalent, we show that they have the same truth table.

Examples

1. Show that $\forall P, Q$ we have $(P \wedge Q) \implies (\overline{P} \vee Q)$.

Solution:

P	Q	$\overline{P}$	$P \wedge Q$	$\overline{P} \vee Q$	$(P \wedge Q) \implies (\overline{P} \vee Q)$
1	1	0	1	1	1
1	0	0	0	0	1
0	1	1	0	1	1
0	0	1	0	1	1

2. Show that $\forall P, Q, R$ the proposition $P \vee (Q \wedge R)$ and the proposition $(P \vee Q) \wedge (P \vee R)$ are equivalent.

Solution:

P	Q	R	$Q \wedge R$	$\mathbf{P} \vee (\mathbf{Q} \wedge \mathbf{R})$	$P \vee Q$	$P \vee R$	$(\mathbf{P} \vee \mathbf{Q}) \wedge (\mathbf{P} \vee \mathbf{R})$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

we notice that the proposition $P \vee (Q \wedge R)$ and the proposition $(P \vee Q) \wedge (P \vee R)$ have the same truth values. So $\mathbf{P} \vee (\mathbf{Q} \wedge \mathbf{R}) \iff (\mathbf{P} \vee \mathbf{Q}) \wedge (\mathbf{P} \vee \mathbf{R})$.

1.4.2 The Direct Proof

To prove $(P \implies Q)$ is true using a direct proof, we suppose that P is true and we show Q is also true based on P .

Example. Show that for $n \in \mathbb{N}$

$$n \text{ is even} \implies n^2 \text{ is even.} \quad (1.1)$$

Solution:

We assume that n is even, i.e. $\exists k \in \mathbb{N}$, such that $n = 2k$ therefore $n^2 = 4k^2$, we put $k_0 = 2k \in \mathbb{N}$ thus $\exists k \in \mathbb{N}$, $n^2 = 2k_0$. Then n^2 is even.

1.4.3 Proof by Contrapositive

To show that the proposition $P \implies Q$ is true it is enough to show that its contrapositive $\overline{Q} \implies \overline{P}$ is true.

Example. Show that for $n \in \mathbb{N}$

$$n^2 \text{ is odd} \implies n \text{ is odd.}$$

Solution:

We use the contrapositive, it suffices to show that

$$n \text{ is even} \implies n^2 \text{ is even.}$$

This implication is proven by the direct proof (see (1.1)).

1.4.4 Proof by Contradiction

To show that the proposition P is true, we assume that its negation $\bar{P}$ is true and we look for a contradiction.

Example. Show that $\sqrt{2}$ is irrational

Solution:

Suppose $\sqrt{2} \in \mathbb{Q}$; then $\sqrt{2} = \frac{a}{b}$; with a and b ($b \neq 0$) two mutually prime integers.

So, $a^2 = 2b^2$; therefore a^2 is even. But we know that (the integer a^2 is even then the integer a is even), then $a = 2k$, with k integer; therefore $4k^2 = 2b^2$; i.e.: $b^2 = 2k^2$ and in the same way we conclude that b is even, therefore 2 divides a and b ; contradiction with a and b are two mutually prime integers. So $\sqrt{2} \notin \mathbb{Q}$.

1.4.5 Proof by case disjunction

Example. Let n be an integer i.e. $n \in \mathbb{Z}$. Show that $n(n+1)$ is even

Solution:

- 1^{st} case: If n is even, we have $n = 2k$, with $k \in \mathbb{Z}$; then $n(n+1) = 2k(2k+1)$ is even.
- 2^{nd} case: If n is odd, we have $n = 2k+1$, with $k \in \mathbb{Z}$; then $n(n+1) = 2(2k+1)(k+1)$ which is even. In both cases $n(n+1)$ is even.

1.4.6 Proof by a Counter-Example

To prove that a proposition of the form:

$$\forall x \in E : P(x),$$

is false, it is enough to find an x_0 in E such that $P(x_0)$ is false, we call this x_0 a counter example.

Example. To show $\forall x \in \mathbb{R}; x \leq x^2$ is false, just take $x_0 = \frac{1}{2}$, we have $x_0^2 = \frac{1}{4}$. Then $x_0 = \frac{1}{2} > x_0^2 = \frac{1}{4}$. Thus, $\forall x \in \mathbb{R}; x \leq x^2$ is false.

1.4.7 Proof by Induction

The proof by induction allows to prove a property with the form $\forall n \in \mathbb{N}; n > n_0; P(n)$ is true for every natural number $n > n_0$. This proof consists three steps

- 1- We verify that $P(n_0)$ is true (n_0 initial value);
- 2-We assume that $P(n)$ is true;
- 3-We show that $P(n+1)$ is also true based on the fact that $P(n)$ is true.

Example. We will prove by induction that $\forall n \in \mathbb{N}^*; 2n \geq n+1$ is true.

- (1) If $n_0 = 1$, we have $P(n_0)$ is true, because $2 \geq 2$;
- (2) We assume that $P(n)$ is true up to n , it means $2n \geq n+1$ is true;
- (3) We have $2n \geq n+1$, so $2n+2 \geq n+3$. The fact that $n+3 \geq n+2$ implies $2n+2 \geq n+2$.

Then $2(n+1) \geq n+2$. Hence $P(n+1)$ is true.

So, $\forall n \in \mathbb{N}^*; 2n \geq n+1$ is true.